# Smart Forecast

## Predicting Student Final Exam Scores From Early Academic Indicators

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CMPT 310 – D200 Introduction to Artificial Intelligence · Spring 2026

## MOTIVATION & GOAL

- Many students struggle academically but are only identified after final grades are released, when it is too late for meaningful intervention.
- Our system aims to address this issue by using machine learning techniques to identify students who may be at risk of poor performance early in a course.
- The AI system will take student-related data as input, such as attendance records, assignment scores, quiz results, and study-related metrics.
- Using linear regression, the system will analyze these features to predict a student's exam score based on continuous behavior in the dataset.
- The output of the system will be a prediction that can help instructors or institutions identify students who may benefit from additional support.

## SYSTEM OVERVIEW

- Used a student exam dataset from Kaggle.
- Loaded, cleaned, and split the dataset into training and testing sets (80/20 split). It includes features such as hours studied, sleep hours, attendance percentage, and previous scores, with exam score as the target variable.
- Exploratory Data Analysis (EDA) was conducted to understand feature relationships, and the results can be seen under the " DATASET & EXPLORATORY ANALYSIS " section below.
- A Linear Regression model was implemented as the baseline.
- To improve performance, two additional models were implemented: Random Forest Regressor and Support Vector Regression (SVR).
- The "MODEL RESULTS" section provides information regarding all three models.

## AI PIPELINE / SYSTEM ARCHITECTURE

**Student Data**
hours studied
sleep
attendance
previous scores

→

**Preprocess**
drop ID
verify nulls
80/20 split

→

**Explore & Select**
correlation matrix
scatter plots
feature checks

→

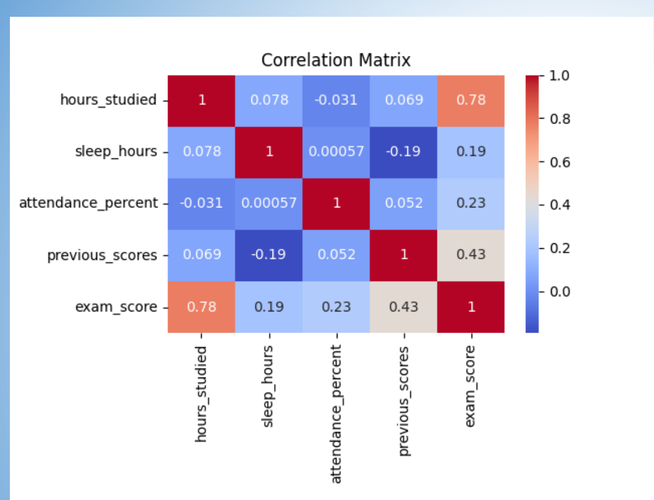
**Train Models**
Linear Regression
Random Forest
SVR + tuning

→

**Predict & Deploy**
best model
score estimate
Flask interface

**Key takeaway: after comparing three regressors under the same evaluation setup, Linear Regression remained the strongest performer ($R^2$ = 0.854, RMSE = 2.79).**

## DATASET & EXPLORATORY ANALYSIS




A heatmap showing correlations between all features and the target variable (exam_score).

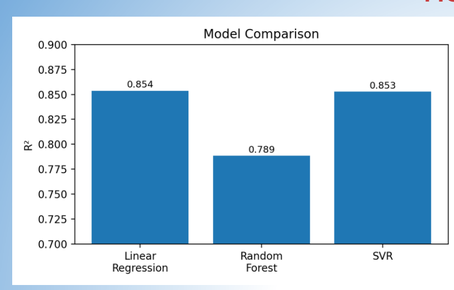
### Analysis:

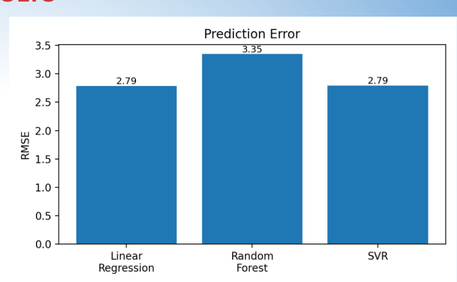
EDA confirmed that the dataset contains meaningful predictive relationships. The strongest correlation with exam score came from hours studied, while previous scores also showed a positive association.

### Explanation:

This visualization demonstrates that meaningful relationships exist in the dataset. For example, strong positive correlations between previous_scores and exam_score confirm that the dataset is suitable for predictive modeling. This supports the validity of the chosen features.

## MODEL RESULTS

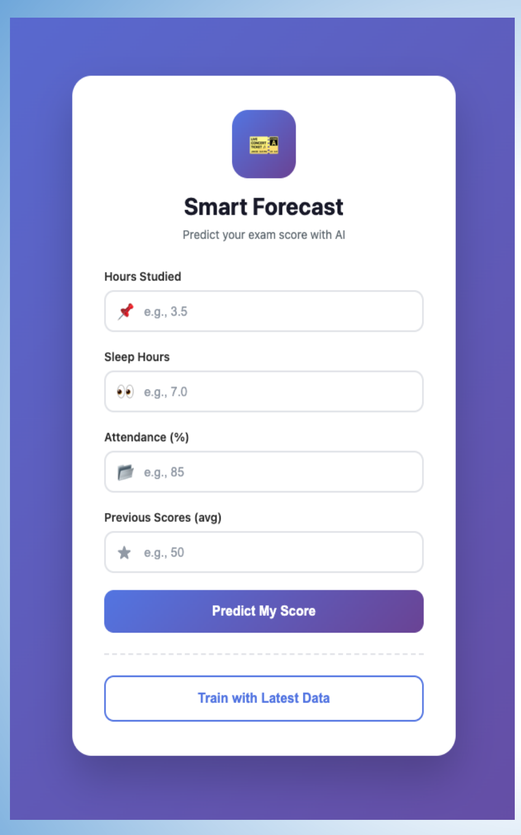



| Model | RMSE | $R^2$ |
|---|---|---|
| Linear Regression | 2.79 | 0.854 |
| Random Forest | 3.35 | 0.789 |
| SVR | 2.79 | 0.853 |

Although Random Forest and SVR were added to capture more complex patterns, the linear baseline stayed the most accurate on this dataset. This suggests the available features already describe the outcome in a fairly linear way.

## FINAL SYSTEM & KEY METHODS AND TECHNOLOGIES USED




Final application

### Why it matters

- supports earlier outreach for students who may need help
- creates a transparent baseline for future educational analytics work
- shows how a small supervised-learning pipeline can move from notebook analysis to an end-user tool

### Current Challenges

- limited feature set constrains what the model can explain
- SVR tuning required scaling and increased computation time
- future work: richer student signals, fairness checks, and broader validation datasets

### Key Methods & Technologies Used

- Python, pandas, NumPy, scikit-learn, matplotlib/seaborn, Flask
- Data cleaning and preprocessing
- Correlation-based EDA and feature analysis
- Linear Regression, Random Forest, SVR
- GridSearchCV hyperparameter tuning
- Evaluation with MSE, RMSE, and $R^2$

## REFERENCES AND HOW TO GUIDE

### References

- Kaggle student exam/performance dataset.
- scikit-learn documentation and library resources
- pandas documentation
- NumPy documentation
- Flask documentation
- Course materials on supervised learning and regression

QR code for Github repository

### How to guide

- Scan the QR code or go to the following link to access our public GitHub repository: https://github.com/RavdeepAulakh/CMPT310SmartForecast.git
- Follow the instructions on How_To_Guide.html to run the application
- Repository also includes a Flask app and training script for a simple browser-based demo.

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